

Faithfulness Under Pressure

Context and History Pruning for Conversational RAG Faithfulness

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Question

When a small language model receives long, noisy retrieved context, can pruning make its answer more faithful to the evidence?

- Task: conversational open-domain question answering with RAG.
- Main benchmark: QReCC, where each question depends on previous conversation turns.
- Output: short answer generated from retrieved/pruned evidence and retained dialogue history.
- Main risk: the model may keep stale history, miss the current evidence, or hallucinate unsupported claims.

Long context is not free.

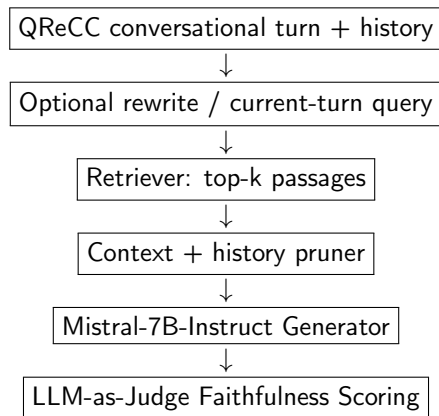
- More tokens increase latency and memory cost.
- Retrieved passages and old dialogue turns often contain near-misses, stale entities, and distractors.
- Small LMs are especially sensitive to noisy evidence.

Source: Liu et al., TACL 2024, “Lost in the Middle”.¹

Long context is not always used well.

- Liu et al. show that models often perform best when relevant information is near the beginning or end of the context.
- Performance can drop when evidence is in the middle, even for long-context models.

¹<https://aclanthology.org/2024.tacl-1.9/>



- The retriever and generator are held fixed across pruning strategies.
- Only retrieved context and retained history change.
- This isolates the effect of pruning on faithfulness, answer quality, and compression.

Pruning Strategies

ID	Strategy	Role
B1	No pruning	Full retrieved context; maximum evidence and maximum noise.
B2	Naive truncation	Hard token budget; common but relevance-agnostic.
B3	RECOMP-style extractive	Selects query-relevant sentences from retrieved passages.
S1	LLMLingua-2	Token-level compression using a bidirectional encoder classifier.
S2	Provenance	Sentence-level sequence-labeling pruner, also related to reranking.
S3	History pruning	Removes redundant context using conversation history.
S4	Combined	Sentence pruning plus history-aware pruning.
S5	CfC/LNN extension	Exploratory adaptive pruning model for multi-turn context.

- **RECOMP**: compresses retrieved documents before generation; reports compression rates as low as 6% with minimal performance loss in evaluated settings.²
- **LLMLingua-2**: formulates prompt compression as token classification and reports $1.6\text{--}2.9\times$ end-to-end latency acceleration at $2\text{--}5\times$ compression.³
- **Provence**: treats context pruning as sequence labeling and reports negligible to no performance drop at almost no additional RAG pipeline cost.⁴

Our angle

We use QReCC because the central problem is conversational: deciding which past turns and retrieved passages still matter for the current question.

²https://proceedings.iclr.cc/paper_files/paper/2024/hash/bda88ed2892f5e61c9a9bf215c566913-Abstract-Conference.html

³<https://huggingface.co/papers/2403.12968>

⁴<https://arxiv.org/abs/2501.16214>

Main benchmark: QReCC

- 14K conversations.
- About 81K question-answer pairs.
- Built for question rewriting in conversational context.
- Retrieval collection: 10M web pages split into 54M passages.
- Each sample includes original question, context, rewrite, answer, and answer provenance.

Current experiment status

- Faithfulness: fraction of generated claims supported by context.
- EM/F1: lexical answer correctness against gold answer.
- Compression ratio: output tokens / input tokens.
- History retention: retained turns / original turns.
- Pilot HotpotQA runs remain only as implementation sanity checks.

Source: Anantha et al., NAACL 2021 / QReCC dataset release.⁵

⁵<https://aclanthology.org/2021.naacl-main.44/>

Pilot Results: HotpotQA Sanity Run

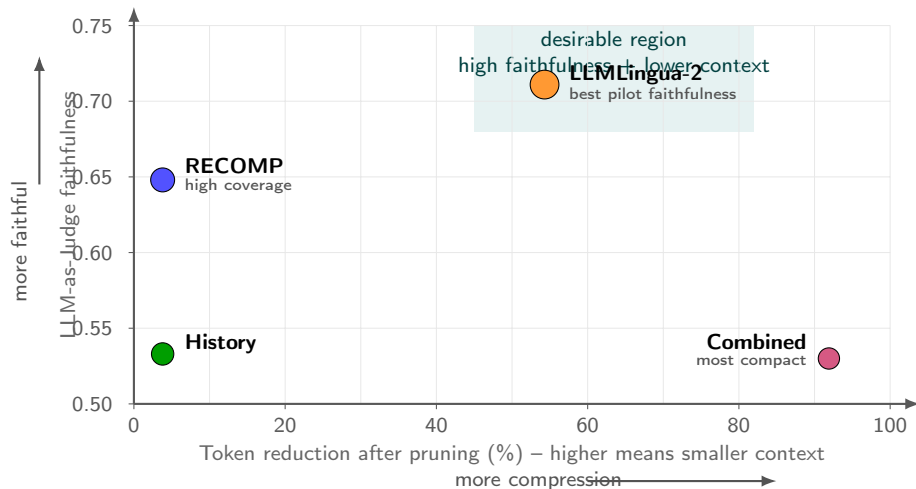
Strategy	Faith.	EM	F1	Comp.	Lat. s
B1: No pruning	0.718	0.040	0.148	1.000	6.76
B2: Naive truncation	0.621	0.040	0.139	0.918	4.95
B3: RECOMP	0.750	0.040	0.141	0.911	5.38
S1: LLMingua-2 fallback	0.750	0.040	0.141	0.911	5.38
S2: Provence	0.657	0.000	0.057	0.072	4.49
S3: History, empty	0.718	0.040	0.148	1.000	6.34
S4: Combined, empty	0.657	0.000	0.057	0.072	4.48

Notebook 2, 50 HotpotQA validation samples. These are pilot implementation checks, not the final QReCC benchmark results.

Strategy	Faith.	EM	F1	Comp.	Cov.	Noise
B3: RECOMP	0.648	0.080	0.195	0.962	0.760	0.624
S1: LLMingua-2 real	0.711	0.040	0.167	0.457	0.000	0.792
S3: History two-turn	0.533	0.100	0.214	0.962	0.627	0.641
S4: Combined two-turn	0.530	0.040	0.112	0.081	0.222	0.284

- LLMingua-2 is the strongest faithfulness result in this pilot run.
- Combined pruning is extremely compact and low-noise, but loses too much support.
- QReCC is the correct next benchmark for testing real history pruning.

Faithfulness–Compression Trade-off



Current numbers are HotpotQA pilot outputs; final benchmark target is QReCC conversational QA.

- **Pruning can improve faithfulness**, but not every pruning method helps equally.
- **Aggressive compression is dangerous for conversational QA**: if the relevant history turn disappears, the current question becomes ambiguous.
- **Faithfulness and correctness are different**: a faithful answer can still be the wrong answer, and a correct short answer can be judged poorly.
- **Exact coverage is too strict for token compression**: LLMLingua-2 may fragment evidence so gold sentences no longer match verbatim.
- **Conversation history is not just extra text**: old turns can help or distract depending on the current question.

Error Analysis Examples

Error type	Example pattern
Evidence removed	The “Arizona” question loses the sentence saying the song was written by Kenny Young.
Verbose correct answer	For yes/no questions, the model may answer correctly with explanation but get EM = 0.
Judge false negative	A short correct phrase such as “Political correctness” can be scored as unfaithful.
Entity near-miss	The generator outputs “T. M. Howard” instead of “T. R. M. Howard”.

Takeaway

We need judge calibration and qualitative analysis before treating one metric as ground truth.

Motivation

Fixed thresholds do not know whether a turn is easy, noisy, redundant, or dependent on earlier dialogue. We want a pruner that adapts over time.

- We explore continuous-time neural models, especially Closed-form Continuous-time networks (CfCs).
- CfCs model temporal dynamics without a heavy numerical ODE solver.⁶
- In our setting, each utterance or sentence receives an importance score.
- The hidden state carries information across conversation turns.

⁶<https://www.nature.com/articles/s42256-022-00556-7>

Idea: surprising utterances deserve more processing time.

For utterance u_i , compute surprisal using a proxy LM:

$$S(u_i) = -\frac{1}{N} \sum_{j=1}^N \log P(w_j \mid w_{<j})$$

Map surprisal to a continuous time step:

$$\Delta t_i = \Delta t_{\min} + \beta S(u_i)$$

- Input: SBERT embedding of each utterance/sentence plus time step.
- Output: importance probability $p_i \in [0, 1]$.
- Teacher labels: leave-one-out impact on Mistral-7B output.
- Training loss: mean squared error between predicted and teacher importance.

Baselines

- Full context.
- Random pruning.
- Cosine-similarity history pruning.
- Existing sentence/token LNN pruners.

Metrics

- ROUGE-L and BERTScore.
- Faithfulness with LLM-as-Judge.
- Token reduction.
- Time to first token.
- History/evidence coverage.

Current status

The CfC/LNN code and Colab notebooks are implemented as an exploratory extension, but final quantitative CfC results are not complete yet.

- Built a modular RAG pruning harness with shared pruner interface.
- Implemented no pruning, truncation, RECOMP-style extraction, LLMLingua-2, Provenance, history pruning, and combined pruning.
- Ran baseline and corrected gap-closing Colab experiments.
- Added coverage/noise metrics to diagnose pruning quality.
- Exported qualitative error-analysis candidates.
- Designed the entropy-driven CfC pruning extension for multi-turn context.

- Complete human annotation and report Cohen's κ for judge calibration.
- Port the final evaluation split from HotpotQA pilot runs to QReCC.
- Tune Provenance and combined-pruning thresholds for conversational evidence coverage.
- Replace exact supporting-sentence coverage with a semantic history/evidence coverage metric.
- Finish CfC teacher-label generation and training.
- Run CfC against full context, random pruning, and cosine-history baselines.
- Evaluate on real QReCC dialogue histories instead of synthetic two-turn histories.

Main finding

Context pruning is useful, but the best pruner is not simply the most aggressive one.

- LLM-Lingua-2 gave the best corrected faithfulness score in our pilot runs.
- Combined pruning showed the strongest compression/noise reduction in pilot runs, but removed too much supporting evidence.
- CfC-style adaptive pruning is our next step for deciding *when* context should be preserved, compressed, or forgotten.

- Liu et al. 2024. Lost in the Middle: How Language Models Use Long Contexts.
<https://aclanthology.org/2024.tacl-1.9/>
- Anantha et al. 2021. Open-Domain Question Answering Goes Conversational via Question Rewriting.
<https://aclanthology.org/2021.naacl-main.44/>
- Pan et al. 2024. LLMingua-2. <https://huggingface.co/papers/2403.12968>
- Xu et al. 2024. RECOMP. https://proceedings.iclr.cc/paper_files/paper/2024/hash/bda88ed2892f5e61c9a9bf215c566913-Abstract-Conference.html
- Chirkova et al. 2025. Provence. <https://arxiv.org/abs/2501.16214>
- Hasani et al. 2022. Closed-form Continuous-time Neural Networks.
<https://www.nature.com/articles/s42256-022-00556-7>